

Zihan Zhang

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Education

Ph.D. The Ohio State University , Statistics	Columbus, OH Aug 2026 – present
M.A. Columbia University , Statistics	New York, NY Sept 2024 – Dec 2025
B.S.; University of North Carolina at Chapel Hill , Statistics and Analytics; Mathematics	Chapel Hill, NC
B.A. • Graduated with Distinction • Dean's List	Aug 2021 – May 2024

Research Experience

Columbia University , Independent Researcher Causal Impact of Diabetes on Cognitive Decline • Supervised by Prof. Bryan Keller. • Studied longitudinal causal effects of diabetes on cognition using marginal structural models, IPTW, overlap weighting, and covariate balancing propensity scores. • Presented at the 2025 Deming Conference on Applied Statistics and received the Deming Poster Award.	New York, NY May 2025 – Feb 2026
University of North Carolina at Chapel Hill , Independent Researcher How Gender Affects Professors' Compensation • Supervised by Prof. Jan Hannig. • Applied propensity score matching, regression adjustment, and causal forests to study gender wage inequality in academia. • Resulting work was accepted for publication in Sankhya B.	Chapel Hill, NC Aug 2022 – May 2025
University of North Carolina at Chapel Hill , Independent Researcher Preferential Attachment in Citation Networks • Supervised by Prof. Xiangying (Zoe) Huang. • Developed mathematical and network models for citation dynamics under preferential attachment. • Completed and defended an Honors Thesis.	Chapel Hill, NC Aug 2023 – May 2024
Irving Institute for Cancer Dynamics, Columbia University , Research Assistant • Research Assistant to Dr. Khanh N. Dinh. • Developed simulation and optimal-transport methods for comparing chromosome instability profiles.	New York, NY Jan 2025 – Dec 2025
Teachers College, Columbia University , Research Assistant • Research Assistant to Prof. Bryan Keller. • Worked on causal inference methods and longitudinal education data.	New York, NY Jan 2025 – Dec 2025
Department of Biostatistics, Columbia University , Research Assistant • Research Assistant to Prof. Yuan Zhang and Prof. Tian Gu. • Processed and harmonized large-scale longitudinal cognitive aging datasets.	New York, NY Sept 2024 – Dec 2024

Department of Data and Systems Engineering, University of Hong Kong, Research Assistant

Hong Kong
Feb 2023 – Aug 2023

- Research Assistant to Prof. Man-Chung Yue.
- Studied causal inference, high-dimensional statistics, and sparse and nonparametric models.

Selected Publications

Evaluating Gender Wage Inequality in Academia using Causal Inference Methods for Observational Data

2026

Zihan Zhang and Jan Hannig. Sankhya B.

Professional Experience

SAS Institute, Biological Data Science Analyst

Beijing, China

- Developed convolutional neural network architectures for COVID-19 lung image classification.
- Applied regularization and sampling methods and evaluated predictive performance using multiple metrics.
- Achieved 93% classification accuracy.

June 2024 – Aug 2024

Selected Projects

Simulation Study of Optimal Matching Designs for Randomized Experiments

2025

Simulation study comparing unmatched randomization, greedy matching, and optimal non-bipartite matching.

- Evaluated covariate balance, efficiency, statistical power, bias, variance, and confidence interval coverage.

Awards

Deming Poster Award

Dec 2025

Awarded at the Deming Conference on Applied Statistics for work on the causal impact of diabetes on cognitive decline.

Distinction

2024

University of North Carolina at Chapel Hill.

Skills

Programming: R, Python, Julia, MATLAB, SQL, SAS, STATA, SPSS, Mathematica

Statistical Methods: Causal Inference, Statistical Modeling, Machine Learning, Longitudinal Data Analysis, Simulation

Tools: LaTeX, Git, VS Code, Jupyter Notebook, HPC

Languages

Mandarin Chinese: Native

English: Advanced